

SYLLABUS

DS4150: Computer Vision [3 1 0 4]

Introduction to image processing: Overview and significance of image processing, Components of an image processing system, Image formation principles, Digitization of images, Continuous and discrete image representations,

Basic image processing operations: Neighbours of pixel, Adjacency and connectivity, Regions and boundaries, Distance measures, Arithmetic operations, Logical operations, Geometrical operations, Convolution and correlation operations,

Image pre-processing techniques and feature extraction: Noise reduction and filtering, Smoothing and sharpening, Histogram equalization, Image enhancement techniques, Point detection, Line and edge detection, Colour, texture, shape and structure features, Spatial and frequency domain features, Corner detection, Hough transform

Applications and Fundamentals of Image Processing and Computer

Vision: Applications of image processing-Document image analysis, Biometrics, Object recognition and tracking, medical image analysis, Content-based image retrieval, Video data processing, Multimedia applications, Virtual reality and augmented reality, Fundamentals of computer vision-What is computer vision?

Low-level, Mid-level and High-level computer vision tasks, Overview of computer vision applications. **Techniques and Representations in Computer Vision:** Low-level vision techniques- Edges, Contours, Textures, Shapes, and Colours, Motion Detection, Optical Flow, and Tracking, Local Features and Invariance, Bag-of-words models, Fisher vector, Mid-level representations-Object parts and attributes, Embedding and representing objects, Understanding and organizing parts and structures.

REFERENCE BOOKS

1. David A Forsyth & Jean ponce “Computer vision – A modern approach”, Prentice Hall, Pearson Education India; Edition: Second, ISBN-13: 978-9332550117, 2015.